

Problem

In a balanced three-phase Y-Y system, the source is an *acb* sequence of voltages and

$$\mathbf{V}_{cn} = 120\angle 35^\circ \text{ V rms}$$

rms. The line impedance per phase is $(1 + j2) \Omega$, while the per-phase impedance of the load is $(11 + j14) \Omega$. Calculate the line currents and the load voltages.

Step-by-step solution

Step 1 of 10

Refer to section 12.2 in the textbook.

In the *acb* phase sequence, the voltage $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$, which in turn leads $\mathbf{V}_{bn}$.

Consider the value of $\mathbf{V}_{cn}$.

$$\mathbf{V}_{cn} = 120\angle 35^\circ \text{ V rms}$$

Compare with $\mathbf{V}_p \angle \theta^\circ$.

The values of $\mathbf{V}_p$ and θ° are 120 V rms and 35° respectively.

Step 2 of 10

Calculate the value of phase voltage $\mathbf{V}_{an}$.

$$\mathbf{V}_{an} = \mathbf{V}_p \angle \theta^\circ + 120^\circ$$

Substitute 120 V rms for $\mathbf{V}_p$ and 35° for θ° .

$$\begin{aligned} \mathbf{V}_{an} &= 120\angle 35^\circ + 120^\circ \\ &= 120\angle 155^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage $\mathbf{V}_{an}$ is $120\angle 155^\circ \text{ V}$.

Step 3 of 10

Calculate the phase voltage $\mathbf{V}_{bn}$.

$$\mathbf{V}_{bn} = \mathbf{V}_p \angle \theta^\circ - 120^\circ$$

Substitute 120 V rms for $\mathbf{V}_p$ and 35° for θ° .

$$\begin{aligned} \mathbf{V}_{bn} &= 120\angle 35^\circ - 120^\circ \\ &= 120\angle -85^\circ \text{ V} \end{aligned}$$

Therefore, the value of the voltage $\mathbf{V}_{bn}$ is $120\angle -85^\circ \text{ V}$.

Step 4 of 10

Consider the single-phase equivalent circuit diagram shown in Figure 1.

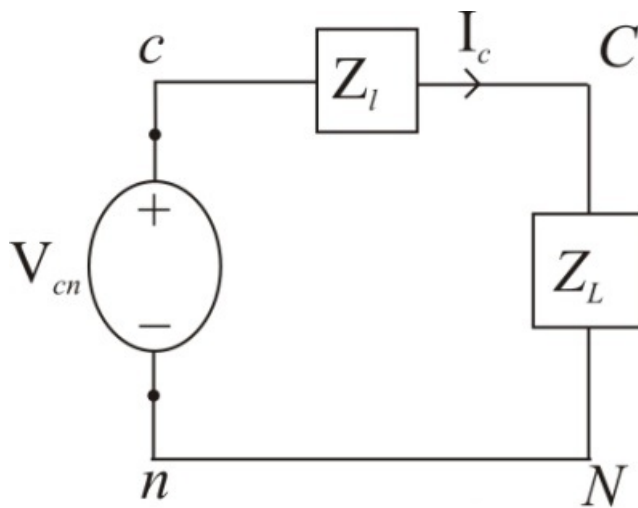


Figure 1

Step 5 of 10

Calculate the value of the current I_c .

$$I_c = \frac{V_{cn}}{Z_l + Z_L}$$

Substitute $120\angle 35^\circ \text{ V rms}$ for V_{cn} , $1 + j2 \Omega$ for Z_l and $11 + j14 \Omega$ for Z_L .

$$\begin{aligned} I_c &= \frac{120\angle 35^\circ}{1 + j2 + 11 + j14} \\ &= \frac{120\angle 35^\circ}{12 + j16} \\ &= \frac{120\angle 35^\circ}{20\angle 53.13^\circ} \\ &= 6\angle -18.13^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_c is $6\angle -18.13^\circ \text{ A rms}$.

Compare $6\angle -18.13^\circ \text{ A rms}$ with $I_p\angle \theta^\circ \text{ A rms}$. The values of I_p and θ° are 6 A rms and -18.13° respectively.

Step 6 of 10

In the acb -phase sequence, the current I_a leads I_c , which intern leads I_b .

Calculate the value of phase current I_a .

$$I_a = I_p\angle \theta^\circ + 120^\circ$$

Substitute 6 A rms for I_p and -18.13° for θ° .

$$\begin{aligned} I_a &= 6\angle -18.13^\circ + 120^\circ \\ &= 6\angle 101.87^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_a is, $6\angle 101.87^\circ \text{ A rms}$.

Step 7 of 10

Calculate the value of phase current I_b .

$$I_b = I_p \angle \theta^\circ - 120^\circ$$

Substitute 6 A rms for I_p and -18.13° for θ° .

$$\begin{aligned} I_b &= 6 \angle -18.13^\circ - 120^\circ \\ &= 6 \angle -138.13^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_b is, $6 \angle -138.13^\circ \text{ A rms}$.

Step 8 of 10

Calculate the load voltage V_{AN} .

$$V_{AN} = I_a Z_L$$

Substitute $6 \angle 101.87^\circ \text{ A rms}$ for I_a and $11 + j14 \Omega$ for Z_L .

$$\begin{aligned} V_{AN} &= (6 \angle 101.87^\circ \text{ A rms})(11 + j14 \Omega) \\ &= (6 \angle 101.87^\circ \text{ A rms})(17.8 \angle 51.84^\circ \Omega) \\ &= 106.83 \angle 153.71^\circ \text{ V rms} \end{aligned}$$

Therefore, the value of load voltage, V_{AN} is $106.83 \angle 153.71^\circ \text{ V rms}$.

Step 9 of 10

Calculate the load voltage V_{BN} .

$$V_{BN} = I_b Z_L$$

Substitute $6 \angle -138.13^\circ \text{ A rms}$ for I_b and $11 + j14 \Omega$ for Z_L .

$$\begin{aligned} V_{BN} &= (6 \angle -138.13^\circ \text{ A rms})(11 + j14 \Omega) \\ &= (6 \angle -138.13^\circ \text{ A rms})(17.8 \angle 51.84^\circ \Omega) \\ &= 106.83 \angle -86.29^\circ \text{ V rms} \end{aligned}$$

Therefore, the value of load voltage, V_{BN} is $106.83 \angle -86.29^\circ \text{ V rms}$.

Step 10 of 10

Calculate the load voltage V_{CN} .

$$V_{CN} = I_c Z_L$$

Substitute $6 \angle -18.13^\circ \text{ A rms}$ for I_a and $11 + j14 \Omega$ for Z_L .

$$\begin{aligned} V_{CN} &= (6 \angle -18.13^\circ \text{ A rms})(11 + j14 \Omega) \\ &= (6 \angle -18.13^\circ \text{ A rms})(17.8 \angle 51.84^\circ \Omega) \\ &= 106.83 \angle 33.71^\circ \text{ V rms} \end{aligned}$$

Therefore, the value of load voltage, V_{CN} is $106.83 \angle 33.71^\circ \text{ V rms}$.